

NASA's Heliophysics Digital Resource Library Digital Librarian Practicum

Isabella Ruiz

Summer, 2026

EDUCATION

MS, Library and Information Science May 2027
The Catholic University of America, Washington, DC

BA, History May 2024
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About

- Passionate about education and information literacy
- Avid reader
- Live music connoisseur
- Star Wars Fan



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Base Model

The SPASE metadata model can be expressed in many forms. One of those forms is as an XML Schema. An XML Schema is a machine readable form (see [schema](#)). Here you can find human readable documentation which includes standard model diagrams.

Current Release is 2.7.1 ([HTML](#) [PDF](#) [XSD](#) [JSON](#)), released: 2025-11-26, [History of changes](#)

SUPPORT

[Email our experts](#) (spase-support@groups.io)

NEED AN UPDATE?

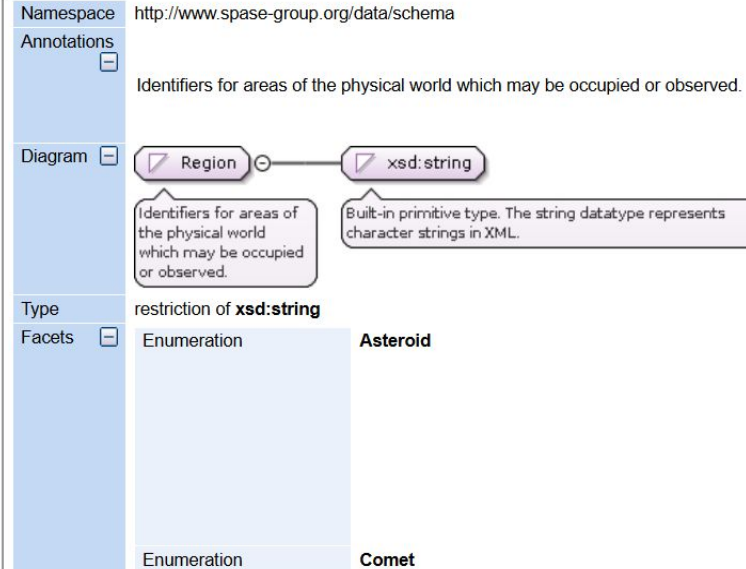
[Request a Change](#)

Project 1

My task was to create code using AI tools to convert each SPASE enumeration list to JSON-LD *without hallucinations*.

Enumeration	Earth.NearSurface.Plasmasphere	A region of the magnetosphere consisting of low energy (cool) plasma. It is located above the ionosphere. The outer boundary of the plasmasphere is known as the plasmopause, which is defined by an order of magnitude drop in plasma density.
Enumeration	Earth.NearSurface.PolarCap	The areas of the globe surrounding the poles and consisting of the region north of 60 deg north latitude and the region south of 60 deg south latitude.
Enumeration	Earth.NearSurface.SouthAtlanticAnomalyRegion	The region where the inner Van Allen radiation belt makes its closest approach to the surface of the Earth. The result is that, for a given altitude, the radiation intensity is higher over this region than elsewhere.
Enumeration	Earth.NearSurface.Stratosphere	The layer of the atmosphere that extends from the troposphere to about 30 km, temperature increases with height. The stratosphere contains the ozone layer.
Enumeration	Earth.NearSurface.SubAuroralRegion	When considering the case of the Earth, the sub-auroral region typically refers to two latitudinal bands, one in the northern

Simple Type `space:Region`



A small extraterrestrial body consisting mostly of rock and metal that is in orbit around the Sun.

Complexity of XSD Structure

colab.research.google.com/drive/1u785EQV-ogyh_H1g0GE6tNjffSLd7gpK?usp=sharing

altLabel of SPASELists.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Now, let's extract the definition for the enumeration list name and store it in a separate dictionary.

```
[1] list_definition_dict = {}

# Ensure cleaned_list_name is available from previous cells
# Ensure json_data is available from previous cells

if 'cleaned_list_name' in globals() and cleaned_list_name:
    if 'list' in json_data and isinstance(json_data['list'], dict):
        dictionary_data = json_data['list']
        if cleaned_list_name in dictionary_data:
            list_overall_data = dictionary_data[cleaned_list_name]
            definition = list_overall_data.get('definition', 'No definition found for this list.')
            list_definition_dict[cleaned_list_name] = definition
            print(f"Extracted definition for '{cleaned_list_name}': {definition}")
        else:
            print(f"'{cleaned_list_name}' not found in the main dictionary.")
    else:
        print("'dictionary' key not found or is not a dictionary in the JSON data.")
else:
    print("No cleaned list name available from previous steps.")

print("\n--- List Definition Dictionary ---")
print(list_definition_dict)
```

Extracted definition for 'Venus': Identifiers for the regions surrounding the planet Venus.

--- List Definition Dictionary ---

{'Venus': 'Identifiers for the regions surrounding the planet Venus.'}

Filtered 1 list Definitions

Variables Terminal

Gemini

Hello, Izzy

How can I help you today?

Save the extracted links to a CSV file

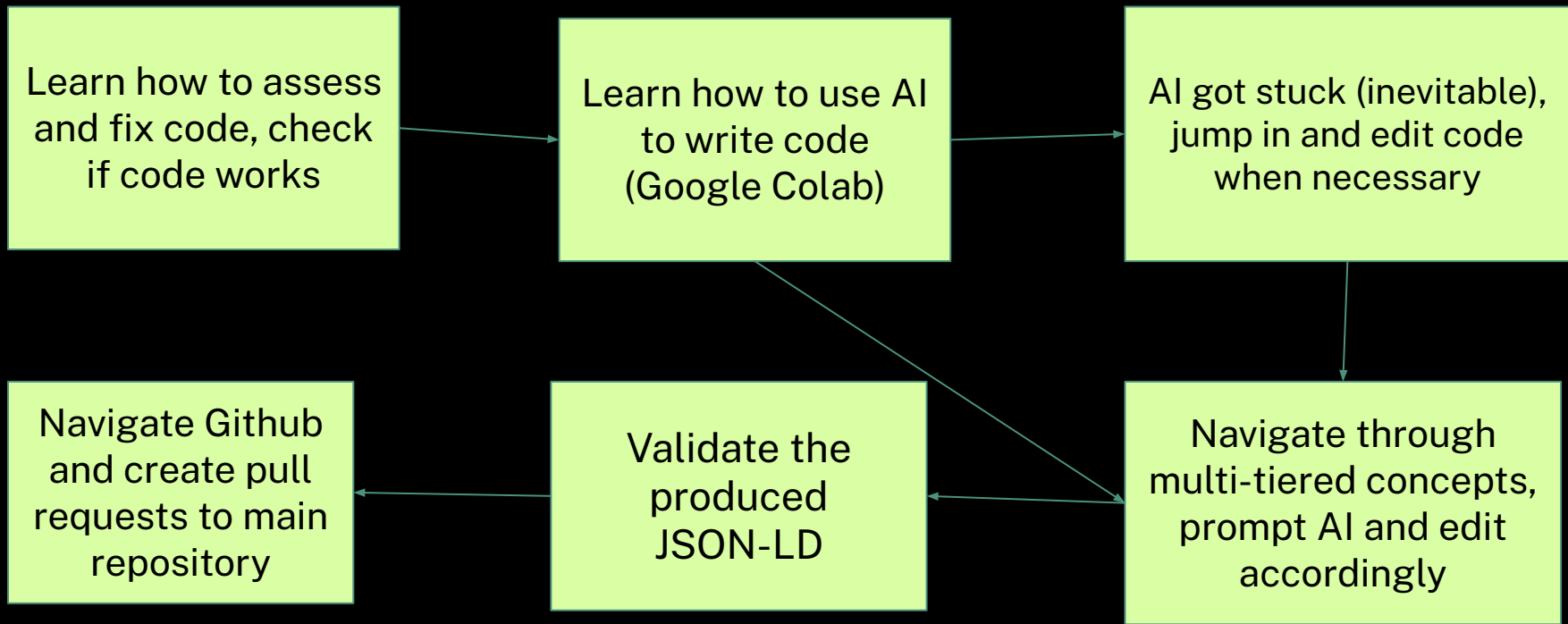
Verify which links are still active using requests

Extract the unique domain names from the list of URLs

What can I help you build?

Gemini 2.5 Flash

Tool Used: Google CoLab/Gemini



Process & Project Workflow

```

    {
      "@type": "Concept",
      "prefLabel": "RadiationBelt",
      "definition": "The region within a magnetosphere where high-energy particles could potentially be trapped in a magnetic field.",
      "inScheme": "https://spase-group.org/data/model/spase-latest/spase-latest_xsd.htm#Region",
      "altLabel": ["Earth.Magnetosphere.RadiationBelt", "Jupiter.Magnetosphere.RadiationBelt", "Mars.Magnetosphere.RadiationBelt", "Mercury.Magnetosphere.RadiationBelt", "Neptune.Magnetosphere.RadiationBelt", "Saturn.Magnetosphere.RadiationBelt", "Uranus.Magnetosphere.RadiationBelt", "Venus.Magnetosphere.RadiationBelt"]
    },
    {
      "@type": "Concept",
      "prefLabel": "RingCurrent",
      "definition": "One of the major current systems confined within planetary magnetospheres. The ring current circles in the magnetic equatorial plane of magnetospheres. It is generated by the longitudinal drift of energetic charged particles trapped on inner, dipole-like magnetospheric field lines. At the Earth, the ring current is carried by 10 keV to 200 keV charged particles typically located at L-shells between 3 and 6. The ring current is also the primary driver of the Sym H and Dst Indices of magnetic storm activity at the Earth.",
      "inScheme": "https://spase-group.org/data/model/spase-latest/spase-latest_xsd.htm#Region",
      "altLabel": ["Earth.Magnetosphere.RingCurrent", "Jupiter.Magnetosphere.RingCurrent", "Mars.Magnetosphere.RingCurrent", "Mercury.Magnetosphere.RingCurrent", "Neptune.Magnetosphere.RingCurrent", "Saturn.Magnetosphere.RingCurrent", "Uranus.Magnetosphere.RingCurrent", "Venus.Magnetosphere.RingCurrent"]
    }
  ]
}

```

Results

<https://github.com/Ruizia-LSC/SPASELists>

In the end, I was able to successfully produce a valid JSON-LD for 68 SPASE enumerations lists. I was also able to upload them to a github repository for use in future projects.

zenodo

Search records...  Communities My dashboard  Log in  Sign up

Planned Intervention: On Thursday, August 13th, 06:15 UTC, Zenodo will be unavailable for 3-5 minutes to perform a storage cluster upgrade.

Published March 28, 2025 | Version v1  Presentation  Open

Improving Research Transparency: An experiment at a NASA SMD Data Repository

Ringuelette, Rebecca ; Thomas, Brian A. ; Stall, Shelley 

Presentation given at the COPDESS workshop on Mar 27, 2025. Presentation shows the results of an experiment performed with a researcher at HDRL (Brian Thomas) on what a researcher is willing to do to incorporate basic open science practices into a publication. The experiment showed that even when a researcher properly creates DOIs for all new materials - data and software in this case - and correctly cites them in their publication, the citation of those materials is sometimes not attributed correctly or at all. The presentation was a conversation starter on what improvements are need in the interlinking infrastructure currently in place.

Files

PublicationTransparency_Mar2025.pdf 

110
 VIEWS

98
 DOWNLOADS

 Show more details

Versions

Version v1	Mar 28, 2025
10.5281/zenodo.15103454	

Cite all versions? You can cite all versions by using the DOI 10.5281/zenodo.15103453. This DOI represents all versions, and will always resolve to the latest one. [Read more.](#)

Task was to build code using AI tools to determine how leaky the citation pipeline is for data and software by comparing the number of citations counted by Sci-X to those properly registered in CrossRef's API calls.

Project 2

Learn how to operate both DataCite's and CrossRef's APIs, read both versions of XML, and understand how to identify multiple types of errors in pipeline.

Task #1

Choose the most important key missing features.

- DOI
- Title
- “Unstructured”

Task #2

Using AI tools, build code to extract API call data for multiple dataset and published DOI's provided by Sci-X and output results into JSON.

Task #3

Build code to produce the data found and visualize how much of the pipeline is missing to track citations to datasets from peer-reviewed publications.

Task #4

Process Overview

<https://github.com/Ruizia-LSC/adsabs-dev-api>



Created by me



1f

Running Extract_DOI_Crossref



Completed - 1w

Github CoPilot Coding Agents

`/home/runner/work/adsabs-dev-api/adsabs-dev-api/cited_list10_0_MassExtractions/Chart_CrossCheckResults.py`

What I changed

- Increased chart readability with larger font sizes across titles, labels, ticks, legends, and annotations.
- **Individual PNG**
 - Pie chart now excludes zero-value categories (so no 0-count search labels or 0% slices are shown).
 - Unstructured chart changed from stacked bar to separate bars (`True` , `False` , `Null`) starting at the x-axis.
 - Added footer metric for:
 - total publication DOIs
 - count where `DOITest=True` and unstructured test is `TRUE` .
- **Comparison PNG**
 - Removed the bar-chart panel.
 - Converted to aligned line-chart layout with shared x-axis positions.
 - Kept x-axis label only on the last line chart.
 - Added y-axis section labels/titles:
 - `DOI and Title Test`
 - `Unstructured Test`
 - Added a third line chart with:
 - success rate: `(DOITest=True or TitleTest=True) / citation_DOIs`
 - SciX DOI detected rate: `citation_DOIs / citation_count`
 - Moved cited_list DOI summary text from top to bottom panel.

Validation

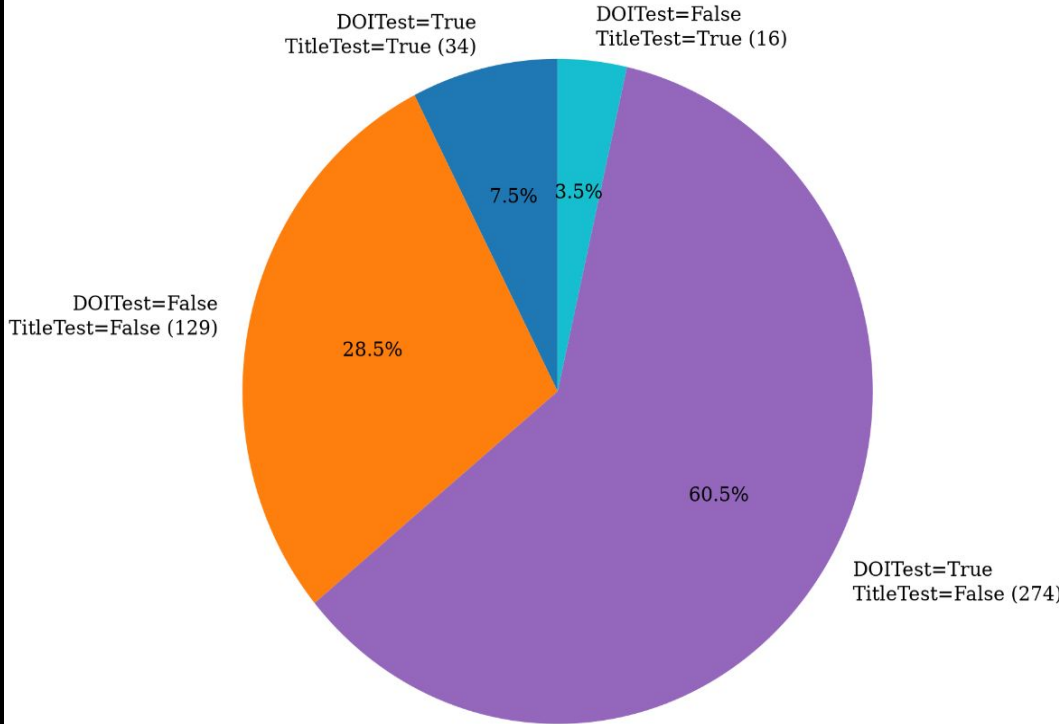
- `python -m py_compile ../Chart_CrossCheckResults.py` ✓
- Script execution attempt failed in this environment due to missing dependency:
 - `ModuleNotFoundError: No module named 'matplotlib'`
- `code_review` run and feedback addressed ✓
- `codeql_checker` run: 0 alerts ✓
- `runtime-tools-secret_scanning` run on changed file: no secrets detected ✓

Example of Produced JSON

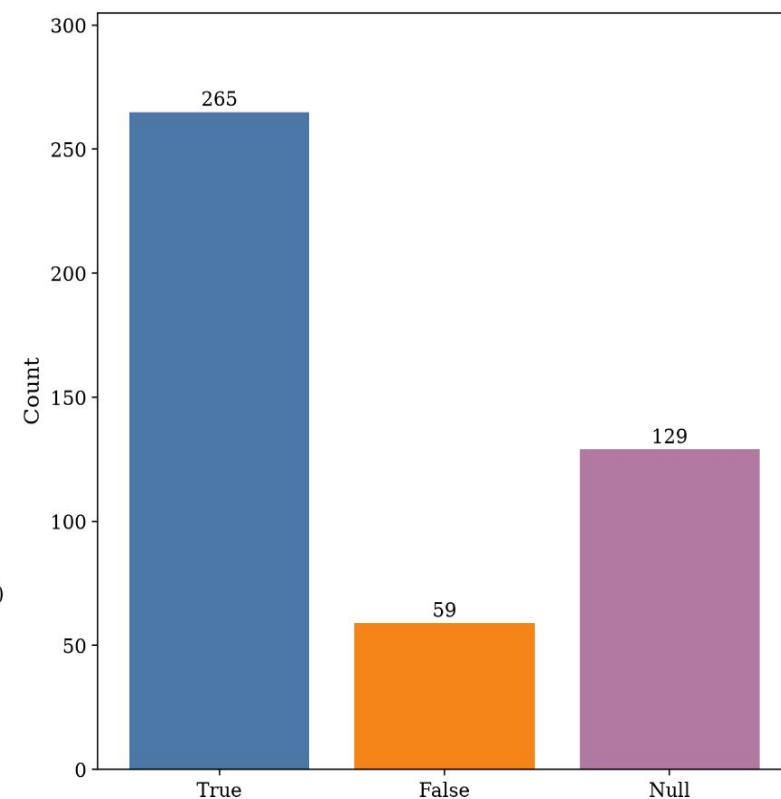


```
{
  "Publication DOI": "10.1016/j.scitotenv.2025.179605",
  "DOI Test": false,
  "Title Test": false,
  "Previous Title": false,
  "Container Found": "null"
},
{
  "Publication DOI": "10.1016/j.scitotenv.2025.180308",
  "DOI Test": true,
  "Title Test": false,
  "Previous Title": false,
  "Container Found": {
    "key": "10.1016/j.scitotenv.2025.180308_bb0120",
    "series-title": "National Land Cover Database (NLCD) (version 3.0) [dataset]. U.S. Geological Survey. doi: 10.5066/P9KZCM54",
    "author": "Dewitz",
    "year": "2024"
  }
},
{
  "Publication DOI": "10.1016/j.scs.2024.105576",
  "DOI Test": true,
  "Title Test": false,
  "Previous Title": false,
  "Container Found": {
    "key": "10.1016/j.scs.2024.105576_bib0012",
    "unstructured": "Dewitz J. (2021). National Land Cover Database (NLCD) 2019 products (ver. 2.0, June 2021) U.S. Geological Survey. 10.5066/P9KZCM54. Accessed March 12, 2023."
  }
},
{
  "Publication DOI": "10.1016/j.srs.2025.100301",
  "DOI Test": false,
  "Title Test": true,
  "Previous Title": false,
  "Container Found": {
    "key": "10.1016/j.srs.2025.100301_bib16",
    "article-title": "National Land Cover Database (NLCD) 2019 Products (ver. 3.0, February 2024)",
    "author": "Dewitz",
    "year": "2021",
    "journal-title": "U.S. Geological Survey Data release"
  }
},
"datasetDOI": "10.5066/p9kzcm54",
"datasetTitle": "National Land Cover Database (NLCD) 2019 Products (ver. 3.0, February 2024)"
```

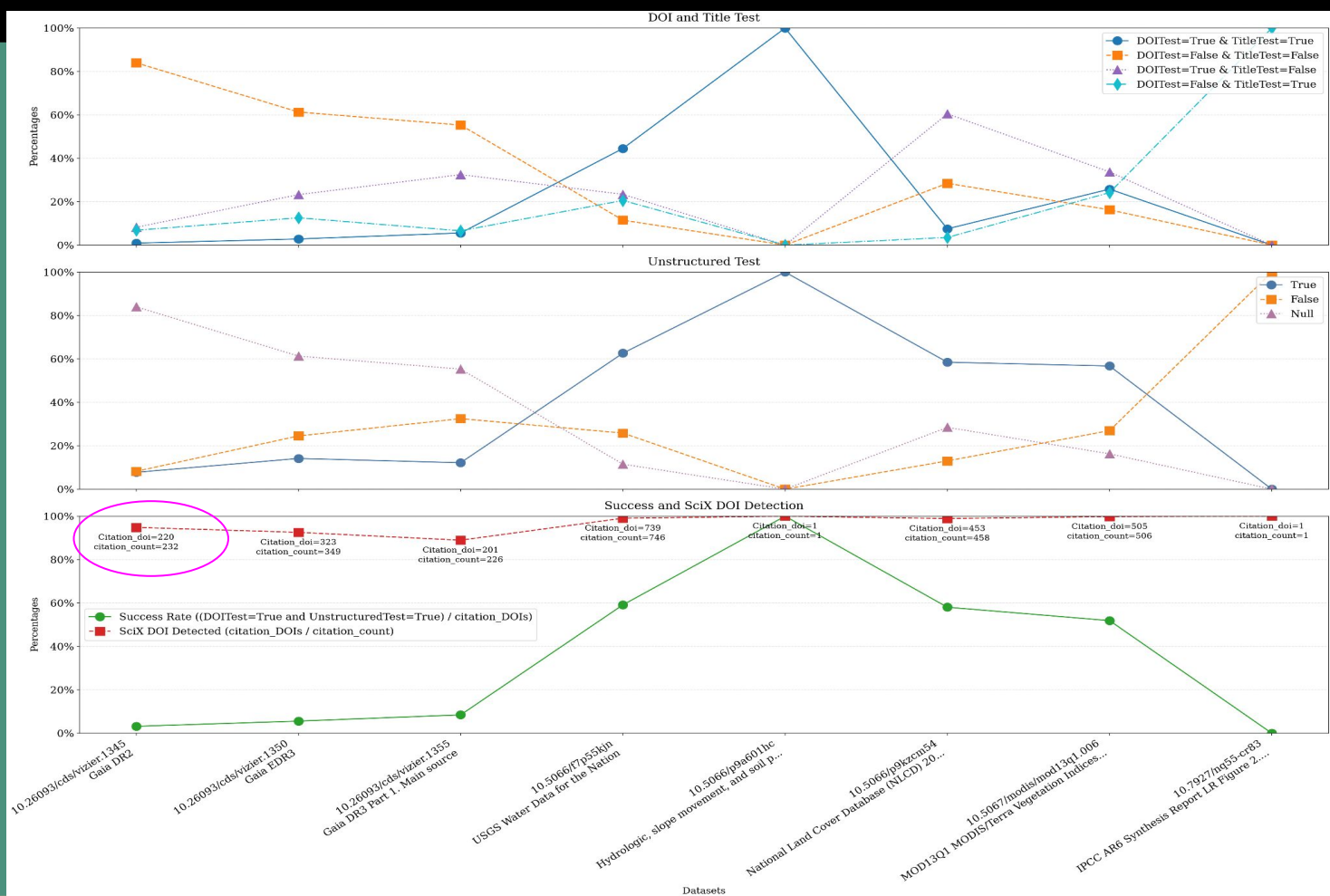
DOITest vs TitleTest



ContainerFound "unstructured" Counts



Total Publication DOIs: 453
DOITest=TRUE and "unstructured"=TRUE: 263



Questions?

